

Formula Sheet Part 2: Asset Pricing, Diagnostics, and EMH

This is a two-column topic split from `All_Formulas_From_Chatnotes.md`.

Conventions

$$t, T, h, k, n, p, q, N$$

Variables: t is a time index; T is sample size or the final observed time; h is a forecast horizon; k is a lag or number of model parameters depending on context; n is a return horizon; p, q are lag orders; N is the number of assets.

$$F_t$$

Variables: F_t is the information set available at time t .

Returns may be written as decimals or percentages. Check the scale before applying VaR or standard deviation formulas.

Asset Pricing Models

CAPM pricing equation

$$\mathbb{E}(R_i) - r_f = \beta_i [\mathbb{E}(R_m) - r_f]$$

Variables: R_i is asset return; R_m is market return; β_i is market beta.

CAPM regression

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i(r_{m,t} - r_{f,t}) + e_{i,t}$$

Variables: $r_{i,t} - r_{f,t}$ is asset excess return; $r_{m,t} - r_{f,t}$ is market excess return; α_i is abnormal return; $e_{i,t}$ is the residual.

CAPM alpha restriction

$$\alpha_i = 0$$

Variables: α_i is abnormal return relative to CAPM.

CAPM total risk decomposition

$$\text{Var}(r_{i,t} - r_f) = \beta_i^2 \text{Var}(r_{m,t} - r_f) + \text{Var}(e_{i,t})$$

Variables: total excess-return variance equals systematic variance plus idiosyncratic variance.

Systematic and idiosyncratic fractions

$$\text{systematic fraction} = R^2$$

$$\text{idiosyncratic fraction} = 1 - R^2$$

Variables: R^2 is the regression explained-variation statistic.

Fama-French three-factor model

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i1}(r_{m,t} - r_{f,t}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + e_{i,t}$$

Variables: β_{i1} is market loading; β_{i2} is SMB loading; β_{i3} is HML loading.

Carhart four-factor model

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i1}(r_{m,t} - r_{f,t}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + \beta_{i4}MOM_t + e_{i,t}$$

Variables: MOM_t is the momentum factor; β_{i4} is momentum loading.

CAPM nested in the four-factor model

$$\beta_{i2} = \beta_{i3} = \beta_{i4} = 0$$

Variables: setting the extra factor loadings to zero reduces the four-factor model to CAPM.

Single-coefficient t-test

$$H_0 : \beta = \beta_0$$

$$H_1 : \beta \neq \beta_0$$

$$t = \frac{\hat{\beta} - \beta_0}{\text{se}(\hat{\beta})}$$

Variables: $\hat{\beta}$ is the estimated coefficient; β_0 is the hypothesised value; $\text{se}(\hat{\beta})$ is the standard error.

Test whether a factor matters

$$H_0 : \beta_{SMB} = 0$$

$$H_1 : \beta_{SMB} \neq 0$$

$$t = \frac{\hat{\beta}_{SMB}}{\text{se}(\hat{\beta}_{SMB})}$$

Variables: β_{SMB} is the SMB coefficient.

Joint test of CAPM against a multi-factor model

$$H_0 : \beta_{SMB} = \beta_{HML} = \beta_{MOM} = 0$$

$$H_1 : \text{at least one extra } \beta \neq 0$$

$$J = \frac{RSS_0 - RSS_1}{RSS_1 / (T - K - 1)}$$

Variables: RSS_0 is restricted-model residual sum of squares; RSS_1 is unrestricted-model residual sum of squares; K is the number of regressors in the unrestricted model.

Residual standard error and RSS

$$RSS = (\text{residual standard error})^2 df$$

Variables: df is residual degrees of freedom.

$$\text{residual standard error} = \sqrt{\frac{RSS}{df}}$$

Variables: residual standard error estimates residual standard deviation.

$$df = T - K - 1$$

Variables: T is sample size; K is the number of regressors excluding the intercept.

Regression Diagnostics and EMH Tests

Regression residual properties

$$\mathbb{E}(e_{i,t}) = 0$$

$$\text{Var}(e_{i,t}) \text{ is constant}$$

$$\mathbb{E}(e_{i,t}e_{i,t-k}) = 0, \quad k = 1, 2, \dots$$

Variables: $e_{i,t}$ is a regression disturbance or residual.

Original model and residual

$$y_t = \alpha + \beta x_t + e_t$$

Variables: y_t is the dependent variable; x_t is the regressor; α, β are coefficients.

$$\hat{e}_t = y_t - \hat{\alpha} - \hat{\beta}x_t$$

Variables: $\hat{e}_t$ is the fitted residual.

Heteroskedasticity hypotheses

$$H_0 : \text{Var}(e_{i,t}) \text{ is constant}$$

$$H_1 : \text{Var}(e_{i,t}) \text{ is not constant}$$

Variables: the null is homoskedasticity; the alternative is heteroskedasticity.

Heteroskedasticity auxiliary regression

$$\hat{e}_t^2 = \gamma_0 + \gamma_1 x_t + \gamma_2 x_t^2 + v_t$$

Variables: γ_j are auxiliary-regression coefficients; v_t is the auxiliary error.

LM statistic for heteroskedasticity

$$W = TR^2$$

Variables: R^2 comes from the auxiliary regression; compare W with a chi-square critical value.

Residual autocorrelation hypotheses

$$H_0 : \text{no autocorrelation}$$

$$H_1 : \text{at least one lagged residual matters}$$

Variables: the alternative means residual serial correlation remains.

Residual autocorrelation auxiliary regression

$$\hat{e}_t = \gamma_0 + \gamma_1 \hat{e}_{t-1} + \gamma_2 \hat{e}_{t-2} + \dots + v_t$$

Variables: $\hat{e}_{t-j}$ are lagged residuals.

LM statistic for autocorrelation

$$AR(p) = TR^2$$

Variables: p is the number of residual lags; under the null compare to χ_p^2 .

HAC definition

HAC = heteroskedasticity and autocorrelation consistent

Variables: HAC/Newey-West changes standard errors, not fitted coefficients.

ETF tracking regression

$$r_{\text{ETF},t} = \alpha + \beta r_{\text{index},t} + e_t$$

Variables: β measures the ETF's exposure to the index.

Leveraged ETF target test

$$H_0 : \beta = -2$$

$$H_1 : \beta \neq -2$$

$$t = \frac{\hat{\beta} - (-2)}{\text{se}(\hat{\beta})}$$

Variables: $\hat{\beta}$ is the estimated tracking slope.

EMH return model

$$r_t = \mu + e_t$$

Variables: μ is constant mean return; e_t is the unpredictable return component.

White noise notation

$$e_t \sim WN(0, \sigma^2)$$

Variables: $WN(0, \sigma^2)$ means white noise with zero mean and variance σ^2 .

White noise conditions

$$\mathbb{E}(e_t) = 0$$

$$\text{Var}(e_t) = \sigma^2$$

$$\text{Corr}(e_t, e_{t-k}) = 0, \quad k \geq 1$$

Variables: no autocorrelation means past shocks do not linearly predict current shocks.

Autocorrelation

$$\rho(k) = \text{Corr}(r_t, r_{t-k})$$

Variables: $\rho(k)$ is autocorrelation at lag k .

Correlogram confidence band

$$\pm \frac{2}{\sqrt{T}}$$

Variables: T is sample size; the band is an approximate 5 percent individual significance guide.

Q-test hypotheses

$$H_0 : \rho_1 = \rho_2 = \dots = \rho_k = 0$$

$$H_1 : \text{at least one autocorrelation is non-zero}$$

Variables: ρ_j is autocorrelation at lag j .

Portmanteau/Ljung-Box statistic from the practice exam

$$Q = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$$

Variables: m is the largest lag tested; $\hat{\rho}_k$ is the sample autocorrelation at lag k ; compare Q with χ_m^2 .

Single autocorrelation large-sample test

$$\sqrt{T} \hat{\rho}(k) \approx \mathcal{N}(0, 1)$$

Variables: $\hat{\rho}(k)$ is the sample autocorrelation at lag k .

$$H_0 : \rho(k) = 0$$

$$H_1 : \rho(k) \neq 0$$

$$|\sqrt{T} \hat{\rho}(k)| > 1.96 \Rightarrow \text{reject at approximately 5 percent}$$

Variables: 1.96 is the two-sided 5 percent normal cutoff.

Demeaned returns for LM predictability tests

$$e_t = r_t - \bar{r}$$

Variables: $\bar{r}$ is the sample mean return.

LM return predictability regression

$$e_t = \gamma_0 + \gamma_1 e_{t-1} + \gamma_2 e_{t-2} + \dots + v_t$$

Variables: lagged demeaned returns test predictability.

LM return predictability hypotheses and statistic

$$H_0 : \gamma_1 = \gamma_2 = \dots = \gamma_k = 0$$

$$H_1 : \text{at least one } \gamma_j \text{ differs from zero}$$

$$AR(k) = TR^2 \sim \chi_k^2$$

Variables: γ_j are lag coefficients; χ_k^2 is chi-square with k degrees of freedom.

Variance-ratio one-period log return

$$r_t = \log(P_t) - \log(P_{t-1})$$

Variables: r_t is one-period log return.

Variance-ratio n-period log return

$$r_t(n) = r_t + r_{t-1} + \dots + r_{t-(n-1)}$$

Variables: $r_t(n)$ is the n -period log return.

Variance scaling under independence

$$\text{Var}[n\text{-period return}] = n \text{Var}[1\text{-period return}]$$

Variables: this holds approximately when returns are independent over time.

Variance ratio

$$VR_n = \frac{\text{variance of } n\text{-period returns}}{n \times \text{variance of 1-period returns}}$$

Variables: $VR_n = 1$ suggests no autocorrelation; $VR_n > 1$ suggests positive autocorrelation; $VR_n < 1$ suggests negative autocorrelation.